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22 January 1988
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MILITARY SPECIFICATION

CANISTER, SMOKE, HC, 105 MM PROJECTILE, M1

This specification is approved for use within U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the filling and assembling for one type of HC smoke canister for a base ejection shell.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals
PPP-B-587 - Boxes, Wood, Wirebound Pallet Type
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-T-76 - Tape, Packaging, Paper (For Carton Sealing)

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-223 - Powder, Black
MIL-A-2550 - Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-650 - Explosive: Sampling, Inspection and Testing
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1168 - Ammunition Lot Numbering
MIL-STD-1235 - Single- and Multi-Level Continuous Sampling Procedures and Tables For Inspection by Attributes

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

TDPL 15-11-22 - Canister, Smoke, HC, 105 MM Projectile, M1

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL 15-11-22 or on the subsidiary drawings shall conform to the specifications listed thereon, unless modified specific characteristics are set forth on the drawings, in which case, detail drawing shall take precedence (see 6.4).

3.1.2 Components. All components of the M1 canister shall conform to the specifications and drawings listed on TDPL 15-11-22 (see 6.4).

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3.2 Filling and assembly. The M1 canister shall be filled and assembled as specified on Drawings 15-11-218 and 15-11-22, respectively (see 6.4 and 6.5). The white smoke mixture (HC) (Drawing 143-1-1) shall be consolidated into the body assembly, after the ignition sleeve, starter slugs/zinc sleeve assembly, and packing disk are in place, in two increments as indicated on Drawing 15-11-218. Each increment shall be consolidated at 39 to 41 tons dead load for a period of not less than 3 seconds. The excess white smoke mixture shall be removed to a faced height as shown on Drawing 15-11-218. Additional packing disks shall be added as required to fill any void. The closing disk shall be installed and the body assembly completed as indicated on Drawings 15-11-218 and 15-11-22, respectively. The packing disk may extend above faced height a distance not to exceed the thickness of the disk.

3.3 Starter slug.

3.3.1 Volatile content. The volatile content of the finished starter slug shall not exceed 0.5 percent when tested in accordance with 4.4.4.2(a).

3.3.2 Crush strength. The crush strength of the starter slug shall be such that the starter slug shall not crush or break when a uniformly distributed load of 500 pounds (lb) is applied parallel to the longitudinal axis of the starter slug and when tested in accordance with 4.4.4.2(b).

3.4 Impregnated sleeve. The impregnated sleeve shall conform to Drawing 15-11-23. After drying, the sleeve shall be ignited and burn over its entire area within 4 seconds maximum when tested in accordance with 4.4.4.2(c).

3.5 Starter mixture. The starter mixture shall conform to the chemical composition specified on Drawing 15-11-119 when tested in accordance with 4.4.4.2(d) and 4.4.4.2(e).

3.6 Functioning. The M1 canister shall produce a continuous dense cloud of white smoke (opaque) for 40 to 95 seconds when tested as specified in 4.4.4.1. Time from first ignition of starter mix to first emission of smoke under pressure shall not exceed 10 seconds. Smoke emission time shall be that period, when smoke is emitted under pressure until the cessation of smoke under pressure. Smoldering of the canister after the charge has burned is not considered part of the smoke emission time. There shall be no interruption of smoke by flame (burning of the starter mixture, first 10 seconds maximum, shall not be construed as flaming).

3.7 First article ballistic information. This specification makes provisions for a first article ballistic informational test prior to commencement of regular production when tested as specified in 4.3.4.

3.8 Preproduction. Prior to the start of regular production, a preproduction lot of canisters shall be produced in accordance with this specification for examination and tests (see 4.3).

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3.9 Workmanship. The canister shall be free from contamination (pyrotechnic mixture, oil, or grease) and damage including but not limited to dents, cracks or deformation which could affect its intended use. The starter slug shall be free from contamination (foreign matter, oil, or grease) and damage including but not limited to cracks, chip outs (see Drawing 15-11-120), dents, or deformation which could affect its assembly into the canister or its intended use. The impregnated sleeve shall be free from contamination (foreign matter, oil, or grease) and damage including but not limited to tears, rips, frayed areas and deformation which could affect its assembly into the canister or its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. The provisions of MIL-A-2550 apply.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Additional responsibility of contractor. Drawings for contractor designed inspection equipment (other than standard measuring equipment) shall be forwarded to design agency for approval prior to production (see 6.2). Ammunition data cards prepared in accordance with MIL-STD-1167 shall be supplied for each lot of cansiters.

4.1.3 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 Preproduction inspection

4.3.1 Lot. A preproduction lot of 1,250 canisters shall be manufactured using the same methods, materials, equipment, processes and cycle time as will be used during regular production.

4.3.2 Sampling.

4.3.2.1 For examination and nondestructive tests.

(a) Examination. Sampling of canisters for examination shall be conducted in accordance with MIL-STD-105.

(b) Filling and assembly. Sampling of canisters for examination in accordance with 3.2 and Drawing 15-11-218 shall be conducted in accordance with MIL-STD-105.

(c) Volatile content of starter slugs. A minimum of two samples of starter slugs shall be selected from each batch (see 6.5) for test in accordance with 4.4.4.2(a).

(d) Impregnated sleeve. A random sample of five each sleeves shall be taken from each batch/lot of sleeves.

4.3.2.2 For destructive tests.

(a) Functioning. Sampling of canisters shall be conducted in accordance with MIL-STD-105, level S-4.

(b) Crush strength of starter slug. Three starter slugs per batch shall be taken at random for the crush strength test in accordance with 4.4.4.2(b).

(c) Impregnated sleeve. A random sample of five each sleeves shall be tested in accordance with 4.4.4.2(c).

(d) Starter mix. Select at random one sample from every 15 batches of starter mixture used to manufacture slugs.

(e) Composite starter mix. Prepare a sample of starter mix composed of material from each batch of starter mix produced during one work shift which shall be blended into one homogeneous sample.

(f) First article ballistic smaple. A random sample of 135 canisters shall be taken from the preproduction lot and submitted to a Government proving ground as designated by the contracting officer (see 6.2).

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4.3.3 Inspection procedure.

4.3.2.1 For examination and nondestructive tests.

(a) Examination. Sample canisters shall be examined in accordance with 4.4.3.3 and MIL-STD-105.

(b) Filling and assembly. Sample canisters of filling and assembly shall be examined in accordance with 3.2 and Drawing 15-11-218 using MIL-STD-105, acceptance number 0. The sample of filled canisters may be examined for length, diameter, and crimp requirements using MIL-STD-105 and the classification of defects (4.4.3.3).

(c) Volatile content of starter slug. The starter slugs shall be tested in accordance with 4.4.4.2(a). The acceptance number is zero.

(d) Impregnated sleeve. Five sleeves shall be examined in accordance with Drawing 15-11-23. The acceptance number is zero.

4.3.3.2 For destructive tests.

(a) Functioning. Sample canisters shall be tested in accordance with 4.4.4.1 and MIL-STD-105. The AQL shall be as specified in 4.4.3.2(a).

(b) Crush strength of starter slug. Three starter slugs per batch shall be tested for the crush strength in accordance with 4.4.4.2(b). The acceptance number is zero.

(c) Impregnated sleeve. Five each sleeves shall be tested in accordance with 4.4.4.2(c). The acceptance number is zero.

(d) Starter mix. The sample starter mix shall be tested in accordance with 4.4.4.2(d). If any sample fails the requirement, a sample from each of the batches represented shall be analyzed. Slugs produced from batches of starter mix found to be defective shall be rejected.

(e) Composite starter mix. The composite starter mix shall be tested in accordance with 4.4.4.2(e). If any sample fails to meet requirements, all the batches produced during that work shift shall be analyzed. Slugs produced from batches of starter mix found to be defective shall be rejected.

(f) First article ballistic functioning. The canister shall be assembled and tested in accordance with 4.3.4. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.3.4 First article ballistic test. The sample canisters taken in 4.3.2.2(f) and received by the proving ground shall be assembled into 45 105 mm, M84A1 smoke projectiles in accordance with Drawing 9223431. Fifteen projectiles shall be temperature conditioned at 70° ± 2.5°F, 15 at -45° + 5°F and 15

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at +145° - 5°F for 12 hours minimum at each temperature. After temperature conditioning, the projectiles shall be fired within 5 minutes of removal from the temperature conditioning at charge zone 7 in a howitzer equipped with a M103 or M137 cannon. The rounds shall be fired at an elevation and fuze setting that will permit ejection of all three canisters in flight and subsequent observation of smoking canisters. The desired canister fall time is 5 seconds, minimum. Appropriate firing tables should be consulted and sufficient stock rounds fired for spotting to verify correctness of elevation and fuze setting. Observation for spotting to verify correctness of elevation and fuze setting. Observation shall be made for the performance of the shell in accordance with table I. In addition, muzzle velocity, chamber pressure, range and time of flight for each round shall be recorded. A smear camera shall be used to observe the projectile in flight at approximately 25 feet in front of the weapon to record any possible malfunction. Copies of the official firing record to include fuze setting and elevation as well as meteorological data shall be sent to the Government agency designated by the contracting officer (see 6.2).

4.3.5 Acceptance/rejection criteria. The preproduction sample shall comply with the examination and tests specified in 4.3.3 to be acceptable. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular produciton.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168, and shall consist of canisters produced in no more than five consecutive work shifts not to exceed 2 weeks consecutive duration. Only one interfix number of conponents from one manufacturer and only one manufacturer's components shall be in one canister interfix number.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests.

(a) Examination. Sampling of canisters for examination shall be conducted in accordance with MIL-STD-105.

(b) Filling and assembly. Sampling of canisters for examination in accordance with the requirements of 3.2 and Drawing 15-11-218 shall be conducted in accordance with MIL-STD-1235, CSP-1, except that sampling for the length, diameter, and crimp requirements for filled canisters may be in accordance with MIL-STD-105.

(c) Volatile content of starter slug. Two starter slugs shall be selected at random from each batch for test in accordance with 4.4.4.2(a).

(d) Impregnated sleeve. A random sample of five each sleeves shall be taken from each batch/lot of sleeves.

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TABLE I. Malfunctions

a.	Misfires. The cartridge fails to fire from the weapon.
b.	Premature functioning and metal parts failure. There shall be no premature burst or indication of metal parts failure in the gun bore or in the first 200 yards (182.88 meters) of horizontal range.
c.	Fuze dud. Absence of visible evidence of fuze functioning as evidenced by a flash or presence of a smoke cloud.
d.	Projectile dud. Canister payload failed to eject.
e.	Lost. Observer(s) fail to see canister ejection or canister burning.
f.	Canister dud. The canister is ejected but no smoking or burning occurs.
g.	Exploder. The canister is ejected and then explodes. This may occur with or without some normal smoke emission. It generally results from smoke mix anomaly and causes a black cloud at the point of incident. The metal parts commonly have the canister lid and/or bottom blown out.
h.	Disintegrator. The canister is ejected and then splits open - usually into approximately equal segments - dispersing the smoke mix into the air. The cloud thus produced is the same color as the fill. The metal parts usually have the canister lid on the flash tube.
i.	Delayed ignition. The smoke emission from the canister did not begin within 12 seconds.
j.	Incorrect burn time. The canister smoke emission time is less than 37 seconds.
k.	Poor smoke. HC smoke that does not obscure.
l.	Flamer. Canister flaming greater than 10 percent of smoke emission time. Canisters that fall in tall, dry grass, will be observed very carefully to insure that the grass does not burn first and then flame the canister.

4.4.2.2 For destructive test.

(a) Functioning. Sampling of canisters shall be conducted in accordance with MIL-STD-105, level S-4.

(b) Crush strength of starter slug. Three starter slugs per batch shall be used for the crush strength test in accordance with 4.4.4.2(b).

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(c) Impregnated sleeve. A random sample of five each sleeves shall be tested in accordance with 4.4.4.2(c).

(d) Starter mix test. Select at random one sample from every 15 batches of starter mixture used to manufacture slugs.

(e) Composite starter mix. Prepare a sample of starter mix composed of material from each batch of starter mix produced during one work shift which shall be blended into one homogeneous sample.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests.

(a) Examination. Sample canisters shall be examined in accordance with the classification of defects and MIL-STD-105.

(b) Filling and assembly. Sample canisters for filling and assembly shall be inspected for requirements of 3.2 and Drawing 15-11-218 using MIL-STD-1235, CSP-1 using an AQL of 1.0 percent defective. The sample of filled canisters may be examined for length, diameter, and crimp in accordance with MIL-STD-105 and classification of defects (4.4.3.3).

(c) Volatile content of starter slug. The volatile content of starter slugs shall be determined in accordance with 4.4.4.2(a). The acceptance number is zero.

(d) Impregnated sleeve. Five sleeves shall be examined in accordance with Drawing 15-11-23. The acceptance number is zero.

4.4.3.2 For destructive tests. Sample canisters shall be tested in accordance with 4.4.4.1 and MIL-STD-105. The AQL shall be as follows:

(a) Functioning. The smoke emission time, delay and a dud* shall have an AQL of 1.0 percent defective.

*A dud is a canister which for any reason fails to function. No canister shall be counted as a defective more than once, regardless of how many defects that one canister exhibits.

(b) Crush strength of starter slug. Three starter slugs per batch shall be used for the crush strength test (4.4.4.2(b)). The acceptance number is zero.

(c) Impregnated sleeve. Five each sleeves shall be tested in accordance with 4.4.4.2(c). The acceptance number is zero.

(d) Starter mix. The sample starter mix shall be tested in accordance with 4.4.4.2(d). If any sample fails the requirement, a sample from each of the batches represented shall be analyzed. Slugs produced from batches of starter mix found to be defective shall be rejected.

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(e) Composite stater mix. The composite starter mix shall be tested in accordance with 4.4.4.2(e). If any sample fails to meet requirements, all the batches produced during that work shift shall be analyzed. Slugs produced from batches of starter mix found to be defective shall be rejected.

4.4.3.3 Classification of defects.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.3.3(a)	Canister, Smoke, HC, 105 MM Shell, M1 Assembly		15-11-22
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:					
101	Marking missing, illegible or incorrect		0.40%	3	Visual
102	Exterior finish missing or incorrect		0.40%	3	Visual
103	Tape missing		0.40%	3	Visual
104	Workmanship		0.40%	3.9	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.3.3(b)	Starter Slug				15-11-120
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:					
101	Inside diameter incorrect		0.40%	3	Gage
102	Outside diameter incorrect		0.40%	3	Gage
103	Length incorrect		0.40%	3	Gage
104	Workmanship		0.40%	3.9	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.3.3(c)	Body Assembly (Filled)		15-11-218
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:					
101	Diameter excessive		0.40%	3	Gage
102	Length excessive		0.40%	3	Gage
103	Crimp incorrect		0.40%	3	Feeler gage
104	Component missing		0.40%	3	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.3.3(d)	Packaging (section 5)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:					
101	Packing wood not preserved or not dry (level A pack only - certification and receiving inspector objective evidence acceptable)		0.40%	5	Certification or receiving inspection
102	Canister packaged incorrectly		0.40%	5	Visual
103	Loose pack		0.40%	5	Visual
104	Component missing		0.40%	5	Visual
105	Strapping missing, loose or incorrect		0.40%	5	Visual/scale
106	Marking illegible or incorrect		0.40%	5	Visual
107	Box damaged or improperly assembled		0.40%	5	Visual

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4.4.4 Tests.

4.4.4.1 Functioning (see 6.4). The test shall be conducted as follows: Remove the squares of tape from the ends and insert a piece of primed time fuze (Bickford type) at least 6 inches long approximately 2 inches in the canister. For priming method see note. Lay canister on its side (flash tube in horizontal position), and ignite the fuze with a fuze lighter or similar device. The canister shall be functioned in an area clear of flammable material, grass, etc., that may influence flaming. If fuze fails to ignite the canister, refuzing will be allowed one time only. If canister fails to light with the second fuze, the canister shall be considered a dud. (Use a stopwatch to check burning times.)

NOTE: Preparation of primed fuze.

A 6 to 8 inch length of safety fuze (class 1375, Bickford type) 0.200 $\pm$ 0.005 inch O.D., burning time 120 seconds per yard $\pm$ 10 percent, shall be primed on one end between 1 and 1/2 inches by dipping in a slurry composed of starter mixture with binder (Drawing 15-11-119) or a slurry composed of black powder (class 5, MIL-P-223) and unplasticized collodion (see 6.6). The fuze may be dipped more than once, depending on the thickness of the slurry, to assure adequate priming. The fuze shall be air dried for a minimum of 2 hours prior to use.

4.4.4.2 Starter slugs.

(a) Volatile content. The volatile content shall be determined for each batch of starter slugs in accordance with MIL-STD-650, method 101.6 Moisture and Volatiles (Vacuum Oven Method).

(b) Crush strength. The crush strength test load on the starter slugs shall be applied uniformly parallel to the longitudinal axis at a constant rate of 1/4 inch per second maximum up to 500 lb. Any crushing or breaking shall be cause for rejection.

(c) Impregnated sleeve. The impregnated sleeve shall be laid on its side clear of flammable material and ignited at one end with a minimum heat source necessary for ignition. Once ignited, the sleeve shall burn as specified.

(d) Starter mix. The sample of starter mix shall be quantitatively and qualitatively analyzed for conformance to Drawing 15-11-119.

(e) Composite starter mix. The composite sample of starter mix shall be qualitatively analyzed for conformance to Drawing 15-11-119.

5. PREPARATION FOR DELIVERY

5.1 Unit packing, level A. Forty-eight canisters of the same type shall be unit packed together in two layers, in a fiberboard box conforming to grade W5c, style RSC, of PPP-B-636. Inside dimensions of the box shall be as

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follows: 18-3/4 inches by 12-1/2 inches by 7-3/4 inches, length by width by depth (l by w by d). Arrangement of the canisters shall be four rows of six canisters in each layer. The layers shall be separated by a fiberboard pad, grade W5c of PPP-F-320, having dimensions of 18-1/2 by 12-1/4 inches. Interlocking partitions shall separate the canisters from one another and shall be of the same material as the fiberboard pads. The partition cell measurements shall be as follows: 3 by 3 by 3-3/4 inches, l by w by d. The fiberboard box shall be closed in accordance with requirements of PPP-B-636, using tape conforming to PPP-T-76.

5.2 Packing.

5.2.1 Level A. Canisters unit packed as specified in 5.1, shall be packed in a wooden shipping container conforming to style 4, class 2, grade A, type 2 load of PPP-B-621. Weight limitations for construction of the shipping container shall be within the 100 to 250 lb category. Fiberboard liners conforming to grade W5c of PPP-F-320 shall be placed within the shipping container to separate the contents from the top, bottom, sides, and ends of the shipping container. Additional fiberboard filler pads shall be used, as required, to provide a tight pack. The inside dimensions of the shipping container shall be 19-1/4 by 13 by 8-3/8 inches, l by w by d. After loading, the shipping container shall be closed and strapped in accordance with the applicable container specification, using flat, galvanized steel strapping conforming to QQ-S-781.

5.2.2 Intraplant shipment. Filled canisters shall be packed in wirebound pallet containers conforming to type I, class 2 of PPP-B-587. Exterior dimensions of the wirebound pallet container shall not exceed 40 by 48 by 50 inches, l by w by d. Canisters shall be uniformly arranged within the container in a manner to maximally utilize all available space. A fiberboard separator pad conforming to grade W5c of PPP-F-320 shall be placed between each layer, in the bottom, and on the top. Additional filler pads conforming to grade W5c of PPP-F-320 shall be utilized to insure a tight pack. The container shall be closed and strapped in accordance with the appendix to PPP-B-587 (see 6.3).

5.2.3 Interplant shipment. For continental U.S. (outside contential U.S. comply with 5.1 and 5.2 - see 6.3), 48 canisters of the same type shall be packed together in two layers in a wooden shipping container conforming to PPP-B-621, style 3, class 1, grade B, type 2 load for domestic shipment. Weight limitations for construction of the box shall be in the 100 to 250 lb category. Case liners and interlocking partitions shall not be required. Inside dimensions of the wood box shall be as follows: 18-1/8 by 12-1/8 by 7-3/4 inches, l by w by d. A separator pad conforming to grade W5c of PPP-F-320 shall separate the top and bottom layers. Fiberboard filler pads conforming to grade W5c of PPP-F-320 shall be used, as required, to assure a tight pack. After loading, the wooden shipping container shall be closed by strapping with two each, 3/8 inch, class C flat steel strapping conforming to QQ-S-781. Nailing of the lids shall not be required.

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5.3 Marking. In addition to any special marking as may be required by the contract or order (see 6.2), wooden boxes shall be marked in accordance with the applicable domestic requirements of MIL-STD-129, except that the crossed retorts insignia shall not be required. The shipping container lid shall be marked with 3/4 inch minimum letters as follows: SMOKE SIGNALS, HANDLE CAREFULLY, KEEP FIRE AWAY - EXPLOSIVE C - HAZARD CLASS 2, CHEMICAL GROUP B.

5.4 Palletization. When specified in the contract or order (see 6.2), shipping containers shall be palletized in accordance with Drawing 15-11-225.

6. NOTES

6.1 Intended use. The HC Canisters are components of the Cartridge, 105 MM Smoke, HC, BE, M84A1 which is used for screening and spotting purposes.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Design agency where drawings for contractor designed inspection equipment are to be forwarded
(c) Government agency when first article sample is to be forwarded
(d) Government agency where official firing record is to be forwarded
(e) Preproduciton:

(1) Time allowed for contractor's submission of samples for Government test and evaluation after award of contract when testing is performed by the Government
(2) Name and address of test facility and shipping instructions when testing is performed by the Government
(3) Time required for the Government to notify the contractor whether to proceed with production

(f) Special marking required (see 5.3)
(g) When palletization is required (see 5.4)

6.3 Interplant shipment. Interplant shipment is packaging for materials that do not enter the military supply system. Typical are shipments from a vendor to a contractor, a subcontractor or a supplier to a military installation or plant.

6.4 Precautions. Standard safety precaution procedures should be taken to prevent fires during preparation of the mixtures and while assembling canisters. Appropriate safety apparel should be worn when handling these materials. Caution should also be exercised during the functioning test as a hazard exists from flying hot particles and possible detonation resulting from incorrect mixture. Manufacture of this item involves pyrotechnic hazards. Potential contractors are advised to be knowledgeable in the art of handling and processing these hazardous items. Also see Edgewood Arsenal Major Component Safety Data Sheet, EADS No. 28.

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6.5 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation to make the final product substantially uniform. A starter mixture batch should not exceed 30 lb.

6.6 Collodion. Unplasticized collodion may be procured from a laboratory supply company.

6.7 Subject term (key word) listing.

Canister
HC
Projectile, 105 MM
Smoke

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1315-A951

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-51360B(EA)	CANISTER, SMOKE, HC, 105 MM PROJECTILE, M1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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